Logical Puzzle Punch Cards

Grade 7: Number Play

Introduction

Solving puzzles always seems complex. But have you wondered how computers solve puzzles? We will see a simplified way to solve some puzzles by following simple instructions and using binary. This will give us insight into how to break complex problems into smaller, easily solvable pieces. By combining smaller problems, computers can solve a larger one.

Activity	Time	Description
Launch	5 min	The teacher will demonstrate by solving a puzzle quickly using the cards, then ask students how it worked. Template: https://tinyurl.com/logiccardstemplate
Exploring Logic punch card by Students	25 min	Students spend time understanding the cards. Then they start solving basic logical statements using the cards. They will then solve some puzzles with multiple statements using the cards. Students can make their own punch cards using the template after class hours. Student Worksheet: https://tinyurl.com/logiccardsworksheet
Discussions & Conclusion	10 min	Discussing the worksheets. The students will continue their discussion of binary numbers from the previous activity. They discuss their understanding of the logical statements and how to deduce from given informations They can also discuss how a complex problem is broken down into individual logical steps and combined to solve a larger problem

CT Components

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Algorithmic Thinking
We follow a step-by-step process to find the required card using the statements provided in the puzzle.

Decomposition
We break the task of solving a logic puzzle into solving individual steps using the binary logic of true or false.

Pattern Recognition
They observe that the number of cards possible is related to the number of holes at the top. They see how true and false, on and off, and U-shaped and O-shaped holes are identical.

Generalisation
From solving a puzzle in 3-holed cards, they see the extension to 5-holed cards.

Logic
These puzzles require solving each step logically.

Activity: Logical puzzle punch cards

In this activity, you will use punched cards to solve logical puzzles. By stacking, aligning, and filtering the cards, we will see how complex reasoning can be built from simple decisions - **True** or **False**, mirroring how binary logic works inside a computer.

The scenario is that you have the following lights in the room: **Yellow Y**, **Pink P**, **Red R**, **Green G**, and **Blue B**. By following the clues given in the **Case Cards**, you have to figure out which lights are **ON** in the given Puzzle. To solve these puzzles, we have a deck of cards representing all possible combinations of these lights being **ON** or **OFF**.

Understanding the cards

To solve these puzzles, we have a deck of cards representing all possible combinations of these lights being **ON** or **OFF**. Each card has five positions for holes on the top, and each position corresponds to a coloured light mentioned above in the given order.

Each position has two possible hole types: **U-shaped** or **O-shaped**. Each card represents one possible combination. If a light is **ON** in a card, then the corresponding hole is **U-shaped**. If a light is **OFF** on a card, then the corresponding hole is **O-shaped**.

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Eg: **Yellow Y** is **OFF**, **Pink P** is **ON**, **Red R** is **ON**, **Green G** is **OFF**, **Blue B** is **OFF** is the card where the holes are **O U U O O**.

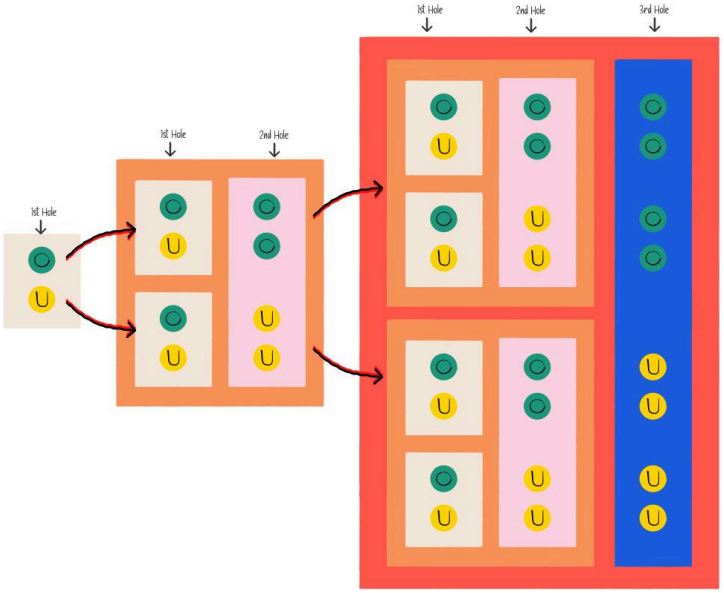


Figure: All the possible three-hole states and how it comes from the two-hole states

Each hole, thus, has 2 possible states. If we have two holes, then we have $2 \times 2 = 4$ possible states. Likewise, if we have 3 holes, we have $2 \times 2 \times 2 = 8$ states. Therefore, if we have 5 holes, we will have $2 \times 2 \times 2 \times 2 \times 2 = 32$ states. That is the number of distinct cards we have.



Figure: Some possible cards with 5 binary digits

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Procedure

If the puzzle says a light is **ON**, we need to remove all cards where that light is **OFF**. For that, we put the stick into the deck of cards through the hole corresponding to that light and pull out all the cards that get stuck to it. We remove these cards because they got stuck due to **O-shaped** holes that indicate the light is **OFF**.

If we know some light is **OFF** in the puzzle, we remove all the cards that do not come out with the stick; these are the cards with **U-shaped** holes in positions corresponding to the given colour, which indicate that the light is **ON**. Ensuring the above helps us solve the logical puzzles through punch cards.



Let's get familiar with the cards and the rules.

1. Can you identify which bulbs are '**OFF**' based on the card?
    a. **Blue B** and **Yellow Y** bulbs
    b. **Green G** and **Red R** bulbs
    c. **Red R** and **Yellow Y** bulbs
    d. All bulbs are OFF

Correct answer: c. **Red R** and **Yellow Y** bulbs

Explanation: The Red R and **Yellow Y** bulbs' positions have **O-shaped** holes. The other bulbs can be seen shining in the image of the card.

Competencies: Logic, Pattern recognition

Understanding using a case

The operation of the cards is explained by showing how they can be used for solving a problem in two-valued logic.



We can start by taking **Case 1**, which contains the first 8 cards from 0 to 7, and focus only on the **Red R**, **Green G**, and **Blue B** lights. Three statements are given containing simple assertions and negation statements. The aim is to determine which bulbs are on or off among these.

To solve this problem, let's establish some notation. Only three terms are involved, so we shall be concerned with only the **Red R**, **Green G**, and **Blue B** holes.

Red bulb '**ON**', 1st position will be **U**	**Red** bulb '**OFF**', 1st position will be **O**
Green bulb '**ON**', 2nd position will be **U**	**Green** bulb '**OFF**', 2nd position will be **O**

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Blue bulb '**ON**', 3rd position will be **U**	**Blue** bulb '**OFF**', 3rd position will be **O**

If all lights are **ON**, then **RGB = U U U**
If all lights are **OFF**, then **RGB = O O O**
If **Red** is **OFF**, **Green** is **ON**, **Blue** is **OFF**, then **RGB = O U O**
If **Red** and **Green** are **ON,** but **Blue** can be anything, then **RGB = U U X**

2. Can you identify which bulb is **OFF** based on this pattern: **RGB = U U O**?
    a. **Red** bulb
    b. **Blue** Bulb
    c. **Green** bulb
    d. All bulbs are OFF

Correct answer: b. **Blue** Bulb
Explanation: The **O-shaped** hole is in the **Blue** bulbs' position
Competencies: Logic, Pattern recognition, Abstraction

Case 1: Focusing on RGB cards only (with 8 cards only)



Statements
I. The **Red** light is **ON**
II. The **Blue** light is **OFF**
III. **ON** light only has **OFF** Neighbour

The problem has three statements.

- *The statement **I**: "The **Red** light is **ON**"*

Insert the stick in the **R** position and lift.
The stick will raise all **RGB = O X X** cards: All cards where **Red** is **OFF**.

RGB = O X X cards are discarded.

All remaining cards are assembled into a pack once more (in any order), and we are ready for the second statement.

3. After performing an action using Statement **I**, how many cards are left in the pack?
    a. 8
    b. 4
    c. 2
    d. 1

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Correct answer: b. 4
Explanation: There are 4 cards with the Red light on and 4 cards with the Red light off. We are keeping the cards which have the light on. The 4 cards with the red light off are removed.
Competencies: Decomposition, Logic, pattern recognition

- *The second statement, **II**, "The **Blue** light is **OFF**"*

Put the stick in the hole corresponding to **Blue** and lift. Cards with **RGB = X X O** will come out. Remaining cards will be **RGB = X X U**.

4. After performing the action using Statement **II**, among these, which card will not be discarded?
    a. **RGB = X X O**
    b. **RGB = U O U**
    c. **RGB = X X U**
    d. All will be eliminated

Correct answer: a. **RGB = X X O**
Explanation: We discard all the cards where **Blue** light is **ON**, **RGB = X X U,** and keep those cards where **RGB = X X O.**

Competencies: Algorithmic thinking, decomposition, pattern recognition

All remaining cards are assembled into a pack once more (in any order), and we are ready for the third statement.



- *The third statement, **III**, "**ON** light only has **OFF** Neighbour"*

Recall the first statement, "The **Red** light is **ON**".
The **Green** bulb neighbours the **Red** bulb. It should be **OFF**.
Insert the stick in the **Green** position and lift. The stick will raise **RGB = X O X** cards.

After this action, only one card will be left, and that will be the solution to the logical puzzle.

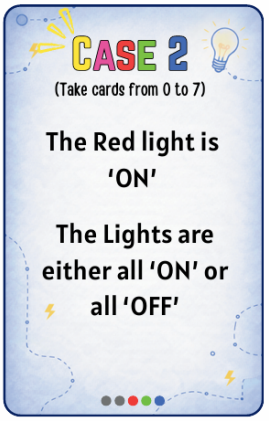
5. What is the pattern of the last card (solution)?
    a. **RGB = U U O**
    b. **RGB = U O U**
    c. **RGB = O U U**
    d. **RGB = U O O**

Correct answer: d. **RGB = U O O**

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Explanation: After the first statement, cards with **RGB = U X X** holes remained. From these, those cards with **RGB = X X O** holes remained after the second statement. Finally, we need the cards with **RGB = X O X** holes. If we put together all this information, we get **RGB = U O O.**

Competencies: Algorithmic thinking, decomposition,



Case 2: Focusing on RGB cards only (with 8 cards only)

Statements
I. The **Red** light is **ON**
II. The Lights are either all **ON** or all **OFF**

The first statement is identical to **Case 1**.
The second statement, **II**, "The Lights are either all **ON** or all **OFF**".

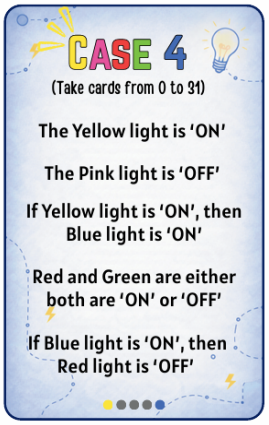
6. Among these, what would be the pattern of the solution?
   a. **RGB = U O O**
   b. **RGB = O O O**
   c. **RGB = U O U**
   d. **RGB = U U U**

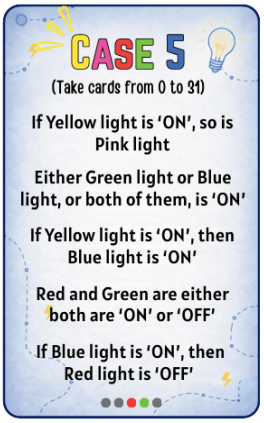
Correct answer: d. **RGB = U U U**
Explanation: The second statement says either all are **ON** or all are **OFF**, but we know from the first statement that **Red** is **ON**. So, all the lights must be **ON.**

Competencies: Logical reasoning, Algorithmic thinking, decomposition

More Puzzles
Below are three progressively harder cases. They introduce additional logic and multi-step filtering to mimic advanced truth-table-based reasoning. Each ends with one solution card.







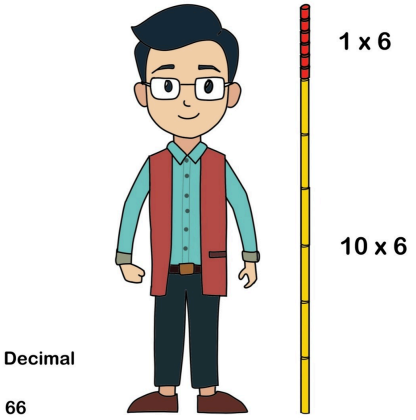
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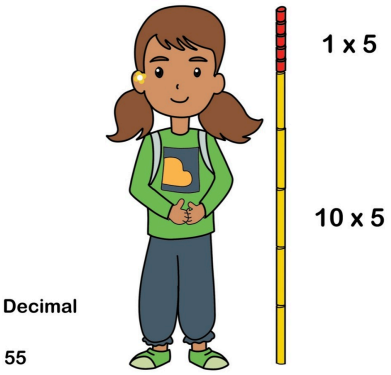
Measure your Height

Grade 8: Story of Numbers

Introduction

We will explore the concept of the positional number system through hands-on activities. We will develop an intuition for various base representations of numbers by measuring students' heights, recognising that decimal numbers that we learn in school are just one way to represent numbers.





Activity	Time	Description
Launch	5 min	The teacher poses the question to the students and invites a couple of students to try their ideas. Activity Video Link: https://tinyurl.com/heightactivityvideo
Problem-solving and algorithm	25 mins	Student's Tryout the activity with each other. Student Worksheet: https://tinyurl.com/heightworksheet As the teacher progresses through the activities, the students progress along with the worksheet.
Discussions and Explorations	5 min	The discussion on the various algorithms, how a slight change in approach leads to a faster solution, and how computers apply such algorithms

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CT Connection

Pattern recognition: Two see that the same patterns of representation appear in each of the base systems and the symbols used in them.

Algorithmic Thinking: We use a step-by-step approach to measure the height using the decimal, binary, and ternary pipes.

Decomposition: Breaking the conversion of the number to binary and ternary into one pipe at a time.

Abstraction: Two use the pattern seen in decimal and apply it to powers of two and powers of three.

Generalisation: Recognising a doubling pattern of 1, 2, 4, and so on, and adding different powers thus generated yields the binary representation; we conclude that any number can be represented in terms of powers of 2. Similarly, we see how to measure using ternary numbers. This method can thus be generalised for any natural number n.

Logic: To understand why the digits start from zero and end before the base number n (0 to $n - 1$)

Activity: Measuring your Height

1: Measuring your height with Decimal Pipes

We generally measure our height in the decimal number system, i.e., using base -10 numbers. We use powers of ten 10^0, 10^1, 10^2, 10^3, … (1, 10, 100, 1000, …) and digits from 0 to 9 (ten distinct symbols), to represent any number that exists.

Imagine now that you are measuring heights using PVC pipes of fixed lengths:
1 inch, 10 inches, 100 inches, and so on
Each pipe size is a power of 10. You also have 9 pipes of each length.



1. How many of the decimal pipes will it take to measure the height of a 66-inch-tall person?

a. 21 pipes: 5 pipes of 10 inches height and 16 pipes of 1 inch height
b. 7 pipes: 7 pipes of 10 inches height
c. 9 pipes: 6 pipes of 10 inches and 3 pipes of 2 inches height
d. 12 pipes: 6 pipes of 1 inch and 6 pipes of 10 inches

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Correct answer: d. 12 pipes: 6 pipes of 1 inch and 6 pipes of 10 inches

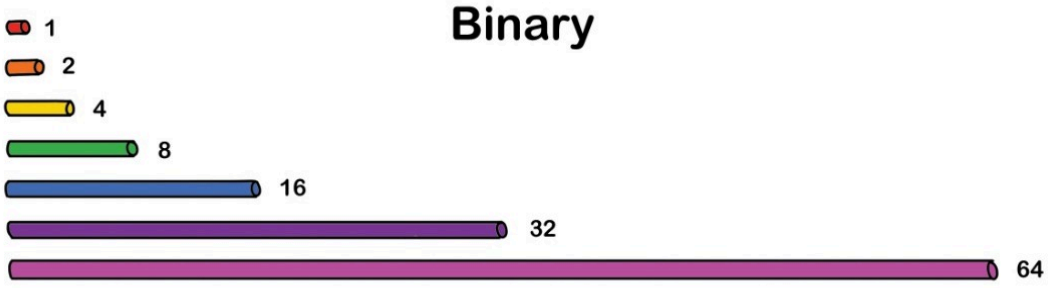
Explanation: 66 in decimal notation is represented as 6x10 + 6x1, that is, 6 of 10s and 6 of 1s, for a total of 12 pipes.

Competencies: Algorithmic thinking, logic, decomposition

2: Measuring your height with binary pipes

We keep hearing that the language of computers is binary, the language of 0 and 1. But what are these binary numbers? How do we understand these in light of what we already know?

Instead of using decimal pipes, which are powers of 10, what happens if we use pipes of length 1, 2, 4, 8, 16, 32, 64, and so on? What is special about these numbers?
They are powers of 2. We have exactly one copy of each pipe length. We will call these pipes Binary pipes.

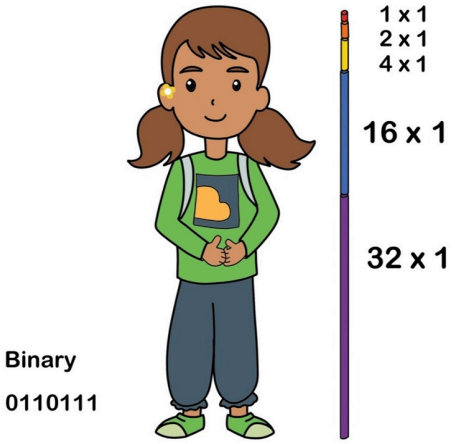


We have only one copy of each pipe for the following reason. Two 2-inch pipes make a 2-inch pipe, but we already have a 4-inch pipe. Two pipes of 4 inches make a 8 inch pipe, but we already have an 8-inch pipe; likewise, we can do this for pipes of all lengths.

Let's measure height in binary. For example, 55 = 32 + 16 + 4 + 2 + 1.
We will see that if we keep a 64-inch pipe, it's more than the height we are measuring, so we don't use a 64-inch pipe. We use the next lower power of 2, 32. We then keep 16, the next lower power. We then keep the next lower power of 2, which is 8, but this is more than the height we are measuring. So we don't use it and move on to the next power of 2. This process continues.

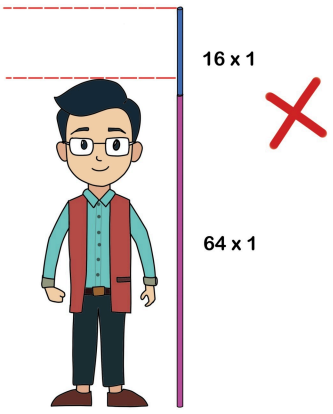
We can tabulate the information below. Using this, we can say that the binary representation of 55 is $(110111)_2$.

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55 in binary	$2^5 = 32$	$2^4 = 16$	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
$(110111)_2$	1	1	0	1	1	1

2. Which all binary pipes will we use to measure the height of a 66-inch-tall person?
    a. 63, 3
    b. 32, 16, 8, 4, 2, 1, 2, 1
    c. 64, 2
    d. 32, 32, 2



Correct answer: c.

Explanation: Let's understand what is happening. We first place a 64-inch-tall block. Now, when we try to keep the other heights over 64 inches, like 32, 16, 8, 4, etc., we will see that these heights will exceed the height we have to measure. So, we pick

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the next lower height until we reach the one that fits. In the number 66, we have 1 copy of 64, 0 copies of 32, 0 copies of 16, 0 copies of 8, 0 copies of 4, 1 copy of 2, and 0 copies of 1. If we write this information in the same way we write place values for decimals, we get:

$2^7 = 128$	$2^6 = 64$	$2^5 = 32$	$2^4 = 16$	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
0	1	0	0	0	0	1	0

Thus, 66 in decimal in binary representation is $(1000010)_2$. We write a $(----)_2$, subscript 2 to tell us that this is a base 2 representation of a number.

Competencies: algorithmic thinking, decomposition, abstraction, generalisation, pattern recognition

3. Binary representation of 10 is:
    a. 1010
    b. 1111
    c. 1000
    d. 1100

Correct answer: a.

Explanation: We can draw a table similar to the one shown above and list the powers of 2 that add up to 10.

2^3	2^2	2^1	2^0
1	0	1	0

Thus, $10 = 1 \times 8 + 1 \times 2 = (1010)_2$.

Competencies: algorithmic thinking, abstraction, generalisation, pattern recognition

4. Why can't we use the digit “2” in the representation of a number in binary?
    a. It is illegal, as only 0 and 1 are given to us in the rules
    b. We can write any number using 0 and 1; there is no need for 2
    c. Computers dislike it
    d. It is too big

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Correct answer: b.
Explanation: If we used the digit “2” in the binary system in the ones place, it would denote 2 x 1 inch = 2 inches. But this is equivalent to using one 2-inch block, which is already present among the height blocks. This means redundancy.

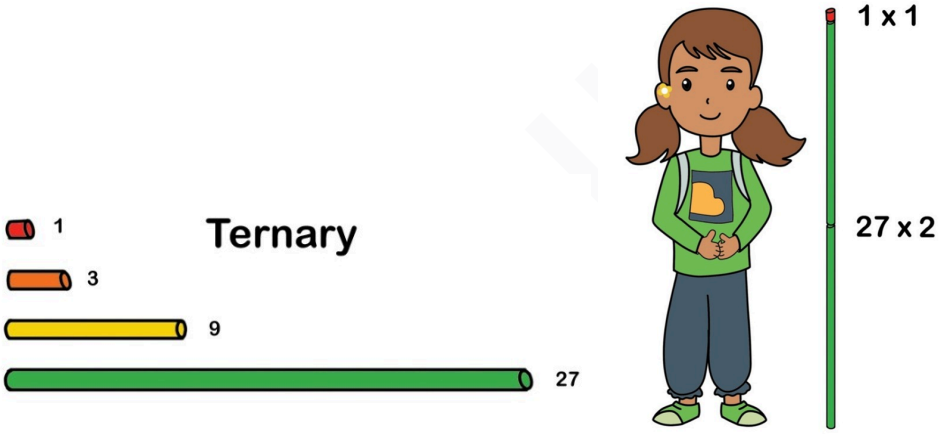
That is why digits must be less than the base. Similarly, two blocks of 4 inches make 8 inches, but we already have an 8-inch block; likewise, two 8 inches is 16 inches, and there is already an 16 inches block

Competencies: abstraction, generalisation, pattern recognition, logic

3: Measuring your height with ternary pipes

Instead of base 10 (decimal) and base 2 (binary), what if we set the base to something else, for example, 3?

Then we will have pipes of lengths 1, 3, 9, 27, 81…, all powers of 3, and 2 pipes of each length.



Let’s see an example of measuring 55 inches in height in ternary.

We will see that if we keep the 81-inch pipe, it's taller than the height we are measuring, so we don’t use it. We use the next lower power of 3, 27. We keep two of these, which is the maximum number of pipes of each type allowed.
We then keep 2 pipes of 9, the next lower power, but this overshoots the height, so we remove one pipe. Even with one 9-inch pipe, we are still over the height we are measuring.
So remove this pipe and keep two 3-inch pipes. But this overshoots the height, so we remove one pipe. Even with one 3-inch pipe, we are still over the height we are measuring.
So, we remove that and keep two pipes of the next lower power of 3, which is 1, but this is more than the height we are measuring. So, we remove one pipe and see that we get the correct height measurement.
55 inches is $54 + 1 = 2 \times 27 + 1 \times 1$.

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55	3^3	3^2	3^1	3^0
$(2001)_3$	2	0	0	1

5. When we measure 66 inches with ternary pipes, how will its representation be?
    a. 2011
    b. 2101
    c. 2110
    d. 2210

Correct answer: c.

Explanation: To measure 66 inches, we use two 27-inch pipes, one 9-inch pipe, one 3-inch pipe, and zero 1-inch pipes.
$66 = 2\times27 + 1\times9 + 1\times3 + 0\times1 = (2110)_3$

66	3^3	3^2	3^1	3^0
$(2110)_3$	2	1	1	0

Competencies: algorithmic thinking, decomposition, abstraction, generalisation, pattern recognition

6. What is the largest digit allowed in Base 3?
    a. 3
    b. 2
    c. 1
    d. 9

Correct answer: b
.
Explanation: As we saw in other representations, the digits allowed in a base number system are numbers less than the base. In case 3, we have three numbers: 0, 1, and 2.

Competencies: abstraction, generalisation, pattern recognition, logic

7. How will we write 100 in base 3 (ternary) notation?
    a. $(3201)_3$
    b. $(2333)_3$

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c. $(11000)_3$

d. $(10201)_3$

Correct answer: d.

Explanation: $100 = 1 \times 81 + 0 \times 27 + 2 \times 9 + 0 \times 3 + 1 \times 1 = (10201)_3$

100	3^4	3^3	3^2	3^1	3^0
$(10201)_3$	1	0	2	0	1

Competencies: algorithmic thinking, decomposition, abstraction, generalisation, pattern recognition

Conclusion

We see that a number can be represented in many forms and symbols. Decimal notation is one of the many possible base number systems (Positional value system).

The fundamental idea is that we arrange the powers of the base number n in increasing order from right to left.

The face value at each place tells us how many times the corresponding power of the base is used to represent the number.

We also see that the base n system requires n distinct digits. These digits are 0 till $n - 1$.

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